Exp 1 : Characteristic study of Side Band Power and Frequencies in AM using MATLAB

Objective 1:

AIM:

To generate an Amplitude Modulated (AM) signal,

perform envelope detection, plot its frequency spectrum,

and calculate the carrier and sideband powers.

Code :

```
clc;
clear;
close all;

Am = 1;
fm = 500;
Ac = 5;
fc = 5000;
fs = 20*fc;

t = 0:1/fs:4/fm;

m = Am*sin(2*pi*fm*t);
c = Ac*sin(2*pi*fc*t);

mu = Am/Ac;

am = Ac*(1 + mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);

dem = abs(hilbert(am));
dem = dem - mean(dem);

N = length(am);
f = linspace(-fs/2, fs/2, N);

AM_FFT = abs(fftshift(fft(am)))/N;

USB = fc + fm;
LSB = fc - fm;
```

```
Pc = Ac^2/2;
Psb = (mu^2*Pc)/4;

figure('Name','AM Modulation','NumberTitle','off');

subplot(3,2,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

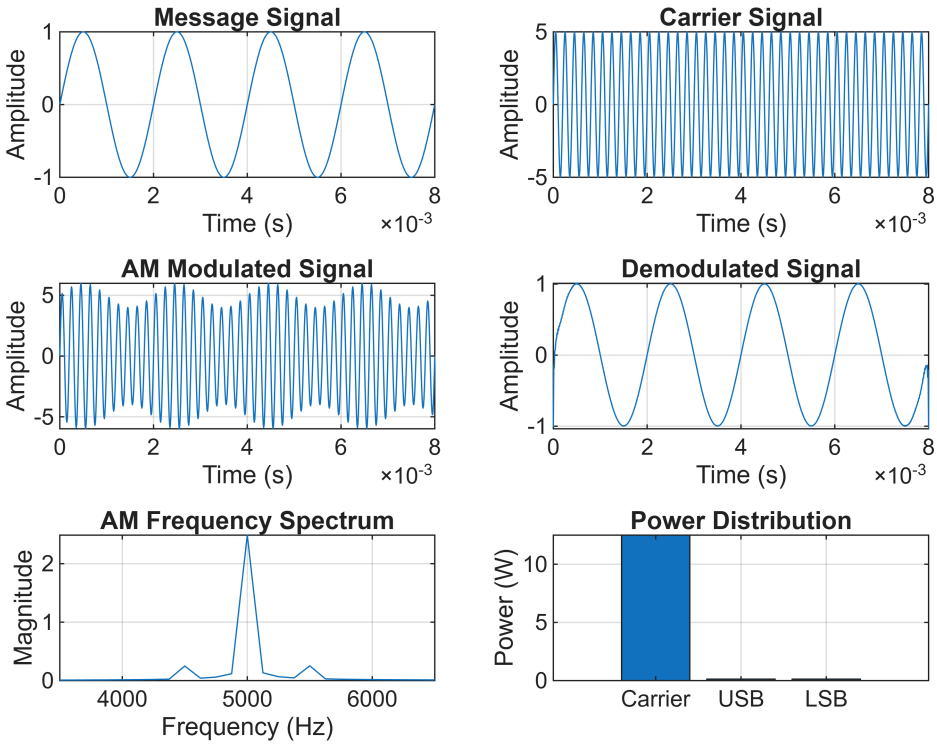
subplot(3,2,2);
plot(t,c);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,2,3);
plot(t,am);
title('AM Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,2,4);
plot(t,dem);
title('Demodulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,2,5);
plot(f,AM_FFT);
xlim([fc-3*fm fc+3*fm]);
title('AM Frequency Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;

subplot(3,2,6);
bar([Pc Psb Psb]);
set(gca,'XTickLabel',{'Carrier','USB','LSB'});
title('Power Distribution');
ylabel('Power (W)');
grid on;
```




```
fprintf('Modulation Index = %.2f\n',mu);
```

```
Modulation Index = 0.20
```

```
fprintf('USB Frequency = %d Hz\n',USB);
```

```
USB Frequency = 5500 Hz
```

```
fprintf('LSB Frequency = %d Hz\n',LSB);
```

```
LSB Frequency = 4500 Hz
```

```
fprintf('Carrier Power = %.2f W\n',Pc);
```

```
Carrier Power = 12.50 W
```

```
fprintf('Power per Sideband = %.3f W\n',Psb);
```

```
Power per Sideband = 0.125 W
```

Objective 2:

To determine and verify the upper sideband(USB) and lower sideband(LSB) frequencies of the AM signal using frequency spectrum analysis in MATLAB

Upper Sideband Frequency (USB)

$$USB = f_c + f_m$$
$$USB = 50 + 5$$
$$USB = 55 \text{ Hz}$$

Lower Sideband Frequency (LSB)

$$LSB = f_c - f_m$$
$$LSB = 50 - 5$$
$$LSB = 45 \text{ Hz}$$

Result:

i) Upper Sideband Frequency (USB) = 55 Hz
ii) Lower Sideband Frequency (LSB) = 45 Hz

Objective 3:

To calculate the carrier power, sideband power, and total transmitted power a given modulation index using MATLAB.

Sideband Power

For an AM signal,

$$P_{SB} = \frac{m^2 P_c}{4}$$

Substituting $m = 1$ and $P_c = 0.5$W:

$$P_{SB} = \frac{(1)^2(0.5)}{}$$

Result :

i) Carrier Power = 0.5 W

ii) Upper Sideband Power = 0.125 W

iii) Lower Sideband Power = 0.125 W

iv) Total Sideband Power = 0.25 W

v) Total Transmitted Power = 0.75 W